# Theory of Automata

Lecture 6

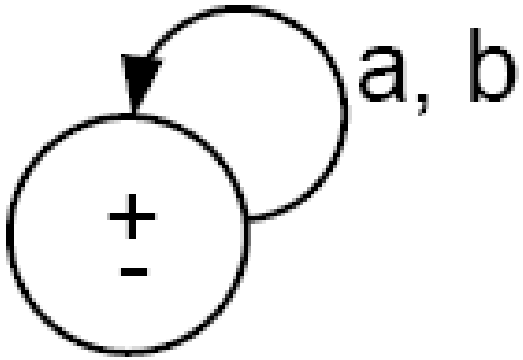
# FA and their Languages

- We will study FA from two different angles:

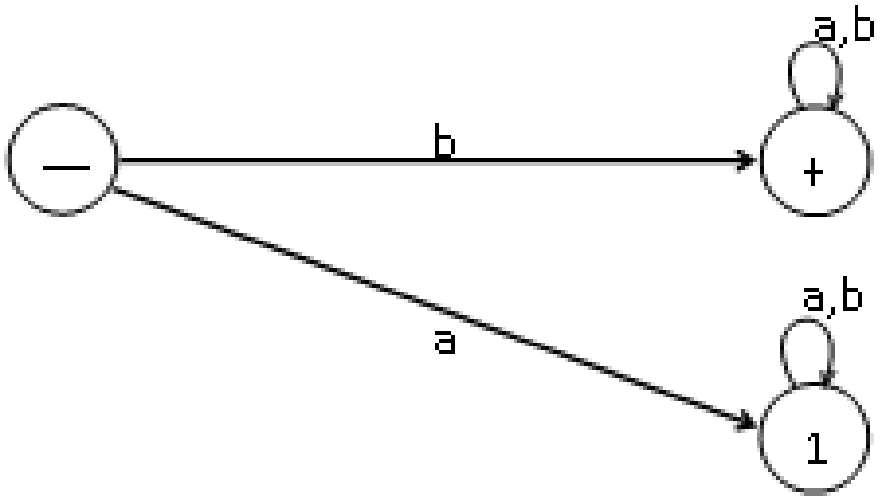
1. Given a language, can we build a machine for it?
2. Given a machine, can we deduce its language?

- **Note:**
  - **Every state has as many outgoing edges as there are letters in the alphabet.**
  - **It is possible for a state to have no incoming edges or to have many.**

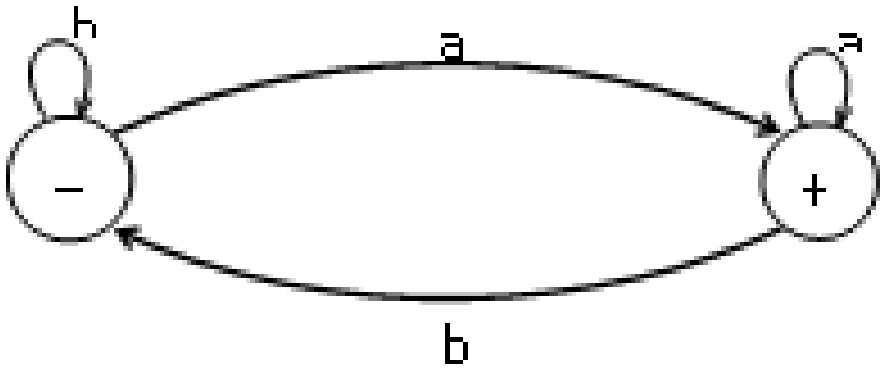
# (a + b)*



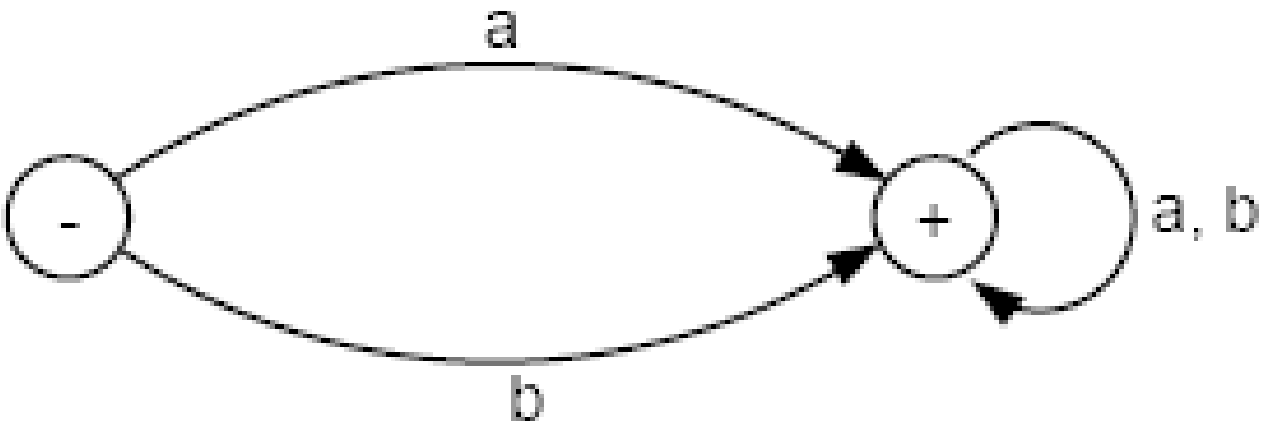
# starting with $b=b(a + b)^*$



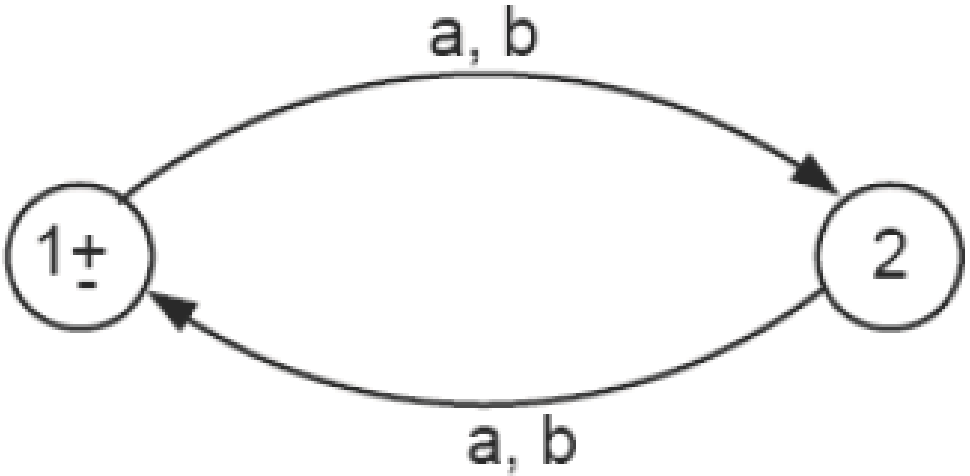
# ending in a=(a+b)*a



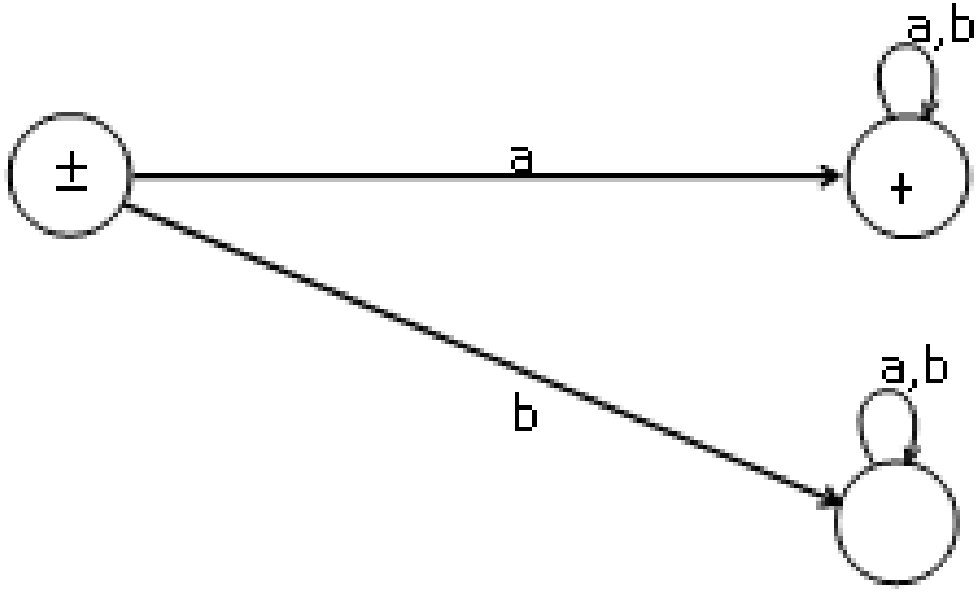
# (a+b)(a+b)*



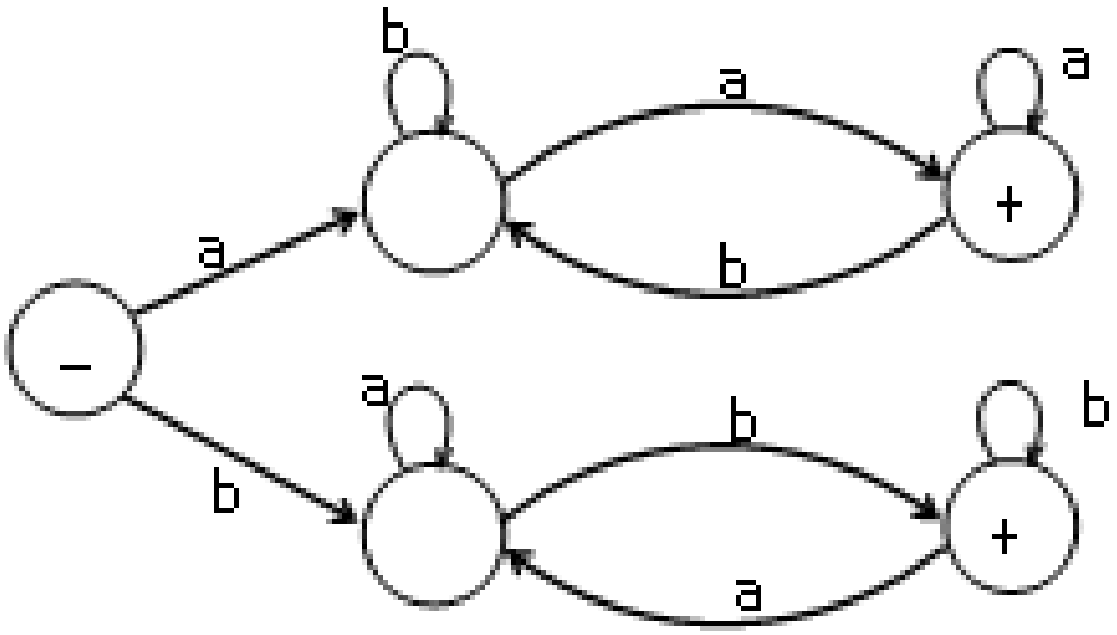
# Even length



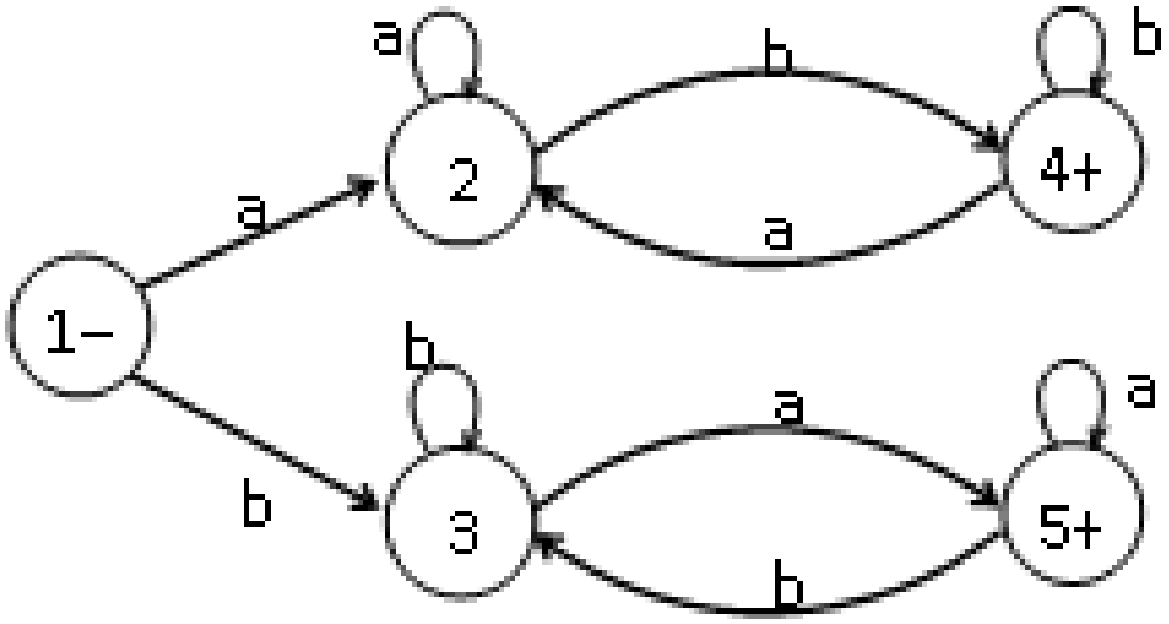
# not beginning with b= a(a + b)* + Λ



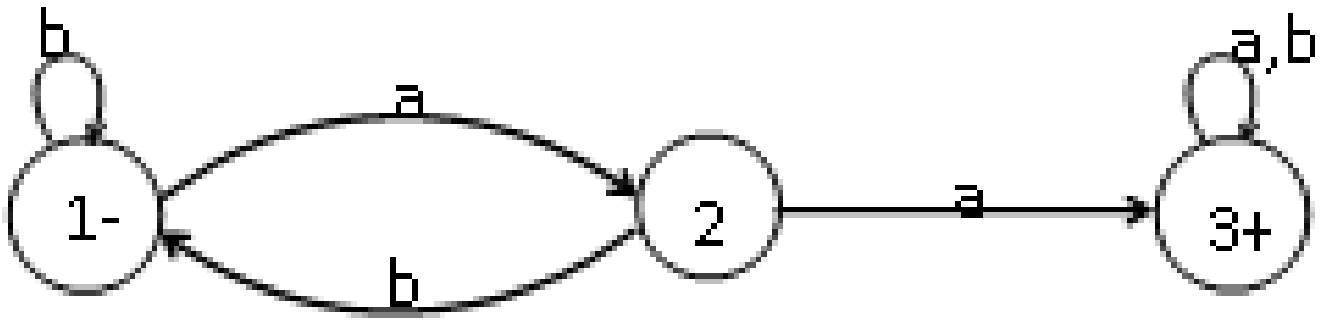
# beginning with and ending in same letters. a(a + b)*a + b(a + b)*b



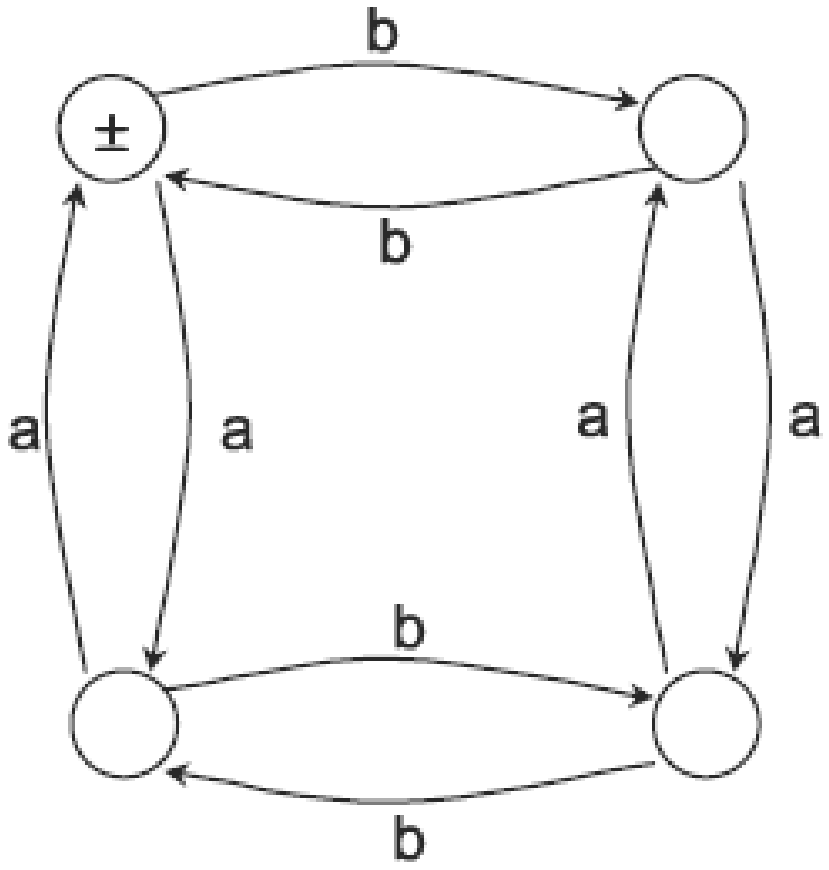
beginning with and ending in different letters. a(a + b)*b + b(a + b)*a



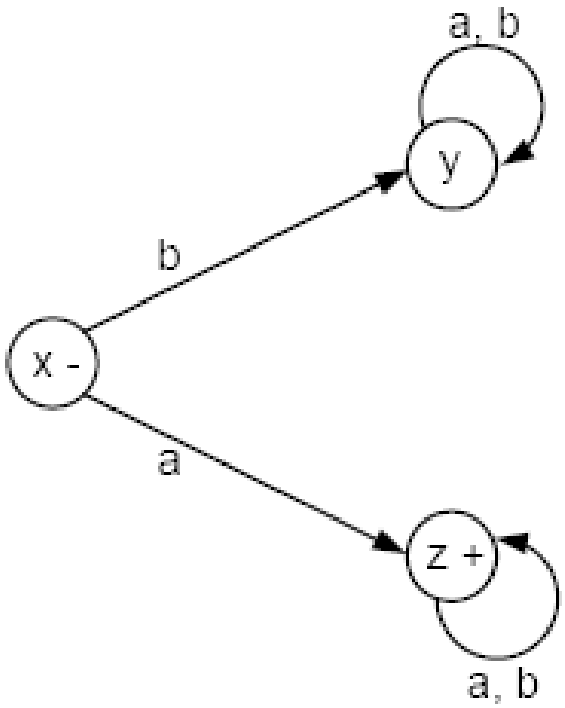
# containing double a=$(a+b)^*$ (aa) $(a+b)^*$.



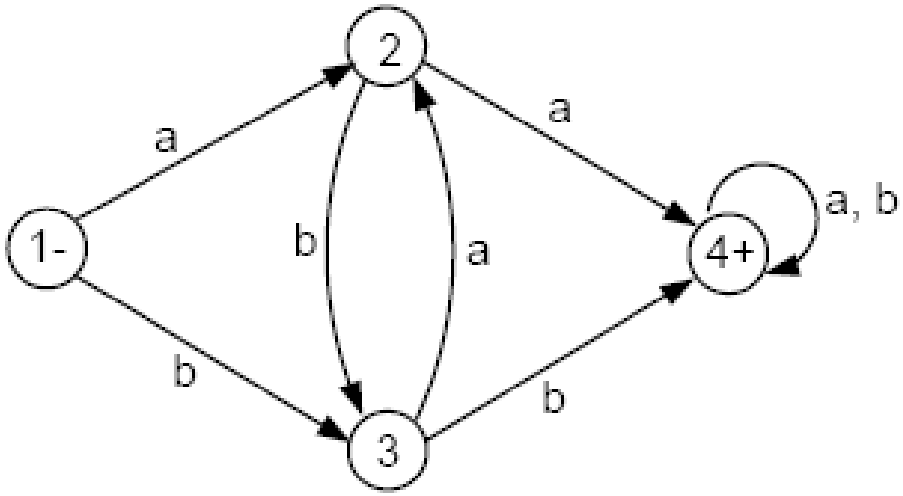
# *EVEN-EVEN*



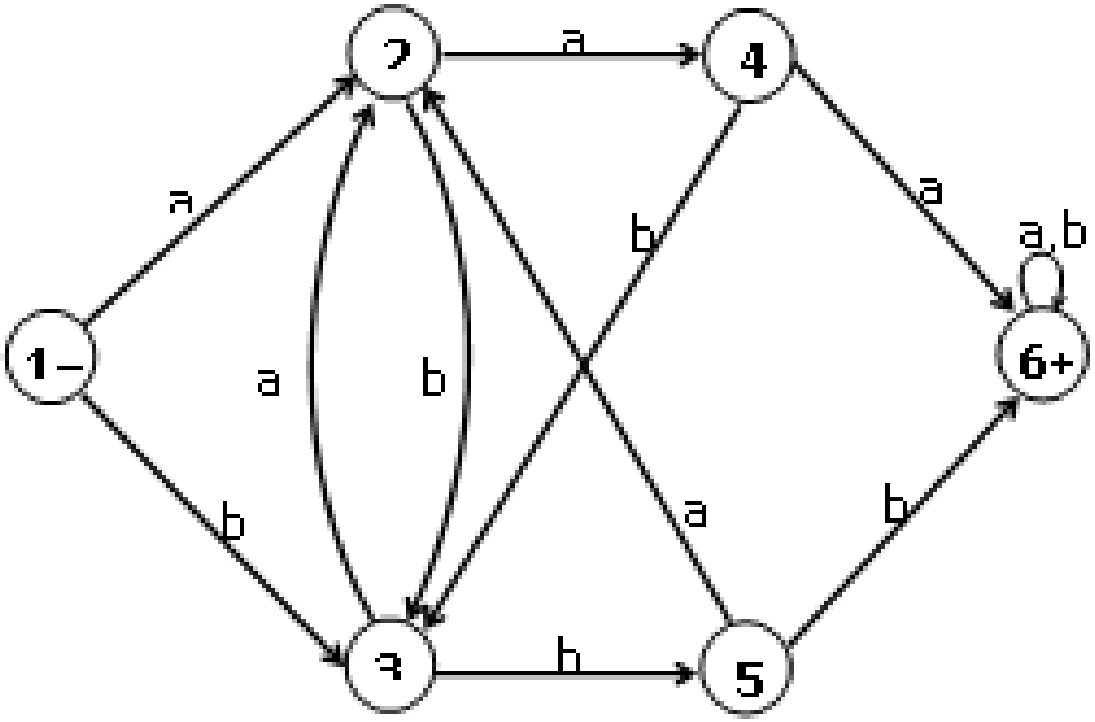
# a(a + b)*



# (a + b)*(aa + bb)(a + b)*

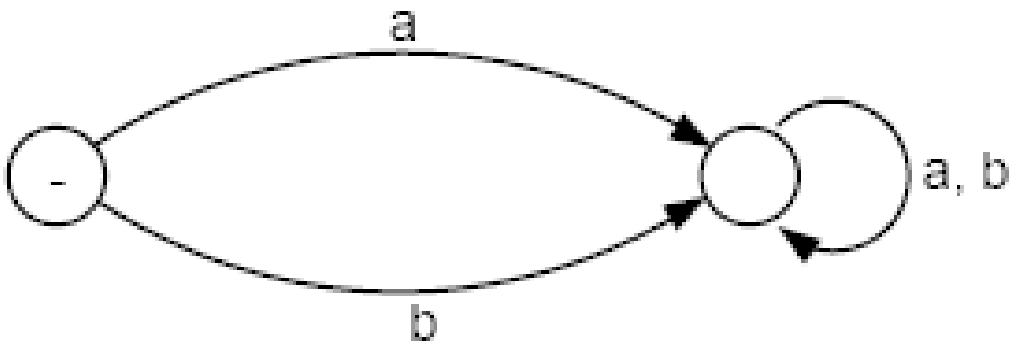


# **containing a triple letter**, either *aaa* or *bbb*



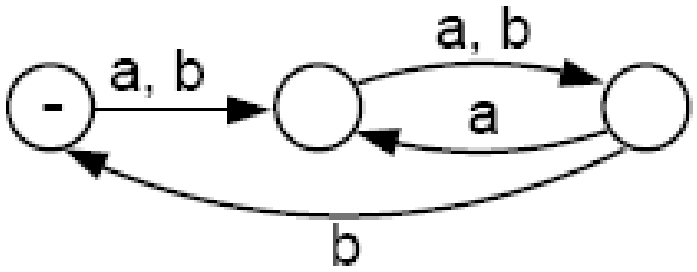
# FAs that accept no language

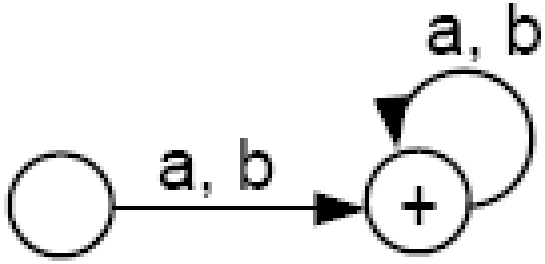
- There are FAs that accept no language. These are of two types:
- The first type includes FAs that have no final states, such as



# FAs that accept no language

- The second type include FAs of which the final states can not be reached from the start state.
- This may be either because the diagram is in two separate components. In this case, we say that the graph is **disconnected**, as in the example below:



- Or it is because the final state has no incoming edges, as shown below:

